

EDS Assignment 2- Formulate 20 problem statements for a given dataset using Numpy and Pandas and Apply Numpy and pandas methods to find the solution for the formulated problem statements.

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link -

<https://colab.research.google.com/drive/1ajj4O8ASR7CVtSs1U8ZdXMPP7DO7WQDo?usp=sharing>

✓ Yelp Dataset Analysis: 20 Problem Statements

This notebook contains 20 analysis tasks performed on the `yelp.csv` dataset using NumPy and Pandas.

Step 1: Uploading the Data

To follow this tutorial, you need the `yelp.csv` file. You can upload it to Google Colab by clicking the folder icon on the left sidebar and then clicking the 'Upload' icon (a page with an up arrow). Once uploaded, the file will be located at `/content/yelp.csv`.

✓ Step 2: Importing Libraries

We need to import `pandas` for data manipulation and `numpy` for mathematical operations. These are the two most common libraries for data analysis in Python.

```
import pandas as pd
import numpy as np

# Load the dataset
df = pd.read_csv('/content/yelp.csv')
display(df.head())
```



	business_id	date	review_id	stars	text
0	9yKzy9PApeiPPOUJEtnvkg	2011-01-26	fWKvX83p0-ka4JS3dc6E5A	5	My wife took me here on my birthday for breakf...
1	ZRJwVLyzEJq1VAihDhYiow	2011-07-27	ljZ33sJrzXqU-0X6U8NwyA	5	I have no idea why some people give bad review...

✓ Problem 1: Basic Dataset Counting

Task: Find the total number of reviews in the dataset and the count of unique business IDs. **Methods used:** `len()` for total count and `nunique()` to find unique values.

```
total_reviews = len(df)
unique_businesses = df['business_id'].nunique()
print(f"Total Reviews: {total_reviews}")
print(f"Unique Businesses: {unique_businesses}")
```

```
Total Reviews: 10000
Unique Businesses: 4174
```

✓ Problem 2: Filtering Rows by Condition

Task: Create a subset of the data containing only reviews that gave exactly 5 stars.

Methods used: Boolean indexing (`df[df['column'] == value]`).

```
five_star_reviews = df[df['stars'] == 5]
display(five_star_reviews.head())
```



	business_id	date	review_id	stars	text	
0	9yKzy9PApeiPPOUJEtnvkg	2011-01-26	fWKvX83p0-ka4JS3dc6E5A	5	My wife took me here on my birthday for breakf...	re
1	ZRJwVLyzEJq1VAihDhYiow	2011-07-27	IjZ33sJrzXqU-0X6U8NwyA	5	I have no idea why some people give bad review...	re
3	_1QQZuf4zZOyFCvXc0o6Vg	2010-05-27	G-WvGalSbqqaMHINnByodA	5	Rosie, Dakota, and I LOVE Chaparral Dog Park!!...	re
4	6ozycU1RpktNG2-1BroVtw	2012-01-05	1uJFq2r5QfJG_6ExMRCaGw	5	General Manager Scott Petello is a good	re

✓ Problem 3: Sorting Data

Task: Sort the reviews to find the top 5 that were marked as most 'useful' by other users.

Methods used: `sort_values()` with `ascending=False` and `head()`.

```
top_useful = df.sort_values(by='useful', ascending=False).head(5)
display(top_useful)
```



	business_id	date	review_id	stars	text
4957	ggLWBFxOKag8dXwVxEvTrg	2011-10-03	1kc50TqtMsIHDGR8yQgF8g	5	Love the place! Amazing Happy Hour Specials
9634	1NZLxU5WvB5roPFzneAILw	2008-09-19	nQHI2kQEiYrMWrpBqdwE4w	4	Scared things in me:\nPar Hilton has a career Why'd th

✓ Problem 4: Creating New Features

Task: Add a new column called `text_length` that stores the character count for every review text. **Methods used:** `apply(len)` to calculate length for every row.

```
df['text_length'] = df['text'].apply(len)
display(df[['text', 'text_length']].head())
```

	text	text_length
0	My wife took me here on my birthday for breakf...	889
1	I have no idea why some people give bad review...	1345
2	love the gyro plate. Rice is so good and I als...	76
3	Rosie, Dakota, and I LOVE Chaparral Dog Park!!...	419
4	General Manager Scott Petello is a good egg!!!...	469

✓ Problem 5: Descriptive Statistics with NumPy

Task: Calculate the average (mean), the middle value (median), and the spread (standard deviation) of review lengths. **Methods used:** `np.mean()`, `np.median()`, and `np.std()`.

```
mean_len = np.mean(df['text_length'])
median_len = np.median(df['text_length'])
std_len = np.std(df['text_length'])
print(f"Mean: {mean_len}, Median: {median_len}, Std: {std_len}")
```



Mean: 710.7387, Median: 541.5, Std: 617.3689564776569

✓ Problem 6: Grouping and Averaging

Task: For each star rating level (1-5), calculate the average number of 'useful', 'funny', and 'cool' votes received. **Methods used:** `groupby()` followed by `mean()`.

```
avg_votes = df.groupby('stars')[['useful', 'funny', 'cool']].mean()
display(avg_votes)
```

	useful	funny	cool
stars			
1	1.604806	1.056075	0.576769
2	1.563107	0.875944	0.719525
3	1.306639	0.694730	0.788501
4	1.395916	0.670448	0.954623
5	1.381780	0.608631	0.944261

✓ Problem 7: Boolean Search

Task: Find all reviews that received more than 10 'cool' votes. **Methods used:** Comparison operator (`>`) inside a filter.

```
cool_reviews = df[df['cool'] > 10]
print(f"Count: {len(cool_reviews)}")
```

Count: 77

✓ Problem 8: Frequency Counts

Task: Count how many reviews exist for every specific star rating (e.g., how many 1-star, 2-star, etc.). **Methods used:** `value_counts()`.

```
print(df['stars'].value_counts())
```

```
stars
4    3526
5    3337
3    1461
```



```
2    927
1    749
Name: count, dtype: int64
```

✓ Problem 9: Conditional Labeling

Task: Create a new column `is_good` that labels a review as 1 if it has 4 or 5 stars, and 0 otherwise. **Methods used:** `np.where()` for conditional logic.

```
df['is_good'] = np.where(df['stars'] >= 4, 1, 0)
display(df[['stars', 'is_good']].head())
```

	stars	is_good
0	5	1
1	5	1
2	4	1
3	5	1
4	5	1

✓ Problem 10: Finding the Maximum

Task: Determine which specific `business_id` appears most frequently in this list of reviews. **Methods used:** `value_counts()` and `idxmax()`.

```
top_business = df['business_id'].value_counts().idxmax()
print(f"Business with most reviews: {top_business}")
```

```
Business with most reviews: JokKtdXU7zXHcr20Lrk29A
```

✓ Problem 11: Working with Dates

Task: Convert the text-based 'date' column into a Python datetime format and extract just the year. **Methods used:** `pd.to_datetime()` and the `.dt.year` accessor.

```
df['date'] = pd.to_datetime(df['date'])
df['year'] = df['date'].dt.year
display(df[['date', 'year']].head())
```



	date	year
0	2011-01-26	2011
1	2011-07-27	2011
2	2012-06-14	2012
3	2010-05-27	2010
4	2012-01-05	2012

✓ Problem 12: Creating a Summary Table (Pivot)

Task: Create a table showing the average 'useful' votes, broken down by year (rows) and star rating (columns). **Methods used:** `pivot_table()`.

```
pivot = df.pivot_table(values='useful', index='year', columns='stars', aggfunc=display(pivot))
```

stars	1	2	3	4	5
year					
2005	NaN	3.000000	2.000000	NaN	3.000000
2006	1.000000	0.333333	1.111111	0.782609	1.285714
2007	1.500000	2.730769	1.730769	2.127119	2.061728
2008	3.564103	2.700000	2.342857	2.708772	3.180995
2009	2.888889	2.220472	1.914773	2.280822	2.458101
2010	1.862903	1.641379	1.283737	1.295833	1.355401
2011	1.361809	1.292135	1.007595	1.075150	1.093348
2012	1.115512	1.074627	0.936709	0.929654	0.911894
2013	0.000000	0.571429	0.333333	0.850000	0.380952

✓ Problem 13: Checking Relationships (Correlation)

Task: Calculate the correlation coefficient to see if longer reviews tend to have higher or lower star ratings. **Methods used:** `corr()`.

```
correlation = df[['text_length', 'stars']].corr()  
display(correlation)
```



	text_length	stars
text_length	1.000000	-0.114674
stars	-0.114674	1.000000

✓ Problem 14: Finding Locations with NumPy

Task: Use NumPy to find the index positions of all reviews that received zero 'funny' votes. **Methods used:** `np.where()`.

```
no_funny_indices = np.where(df['funny'] == 0)[0]
print(f"First 10 indices: {no_funny_indices[:10]}")
```

```
First 10 indices: [ 0  1  2  3  4  7  8  9 11 12]
```

✓ 🍕 Problem 15: Searching Text

Task: Search through all review texts to find rows that mention the word 'pizza'.

Methods used: `str.contains()` with `case=False`.

```
pizza_reviews = df[df['text'].str.contains('pizza', case=False)]
print(f"Reviews mentioning pizza: {len(pizza_reviews)}")
```

```
Reviews mentioning pizza: 683
```

✓ Problem 16: Checking for Missing Data

Task: Verify if there are any empty (null) cells in the entire dataset. **Methods used:** `isnull()` and `sum()`.

```
print(df.isnull().sum())
```

```
business_id    0
date           0
review_id      0
stars          0
text           0
type           0
user_id        0
cool           0
useful         0
funny          0
text_length    0
```




```
is_good      0
year         0
dtype: int64
```

✓ Problem 17: Multiple Filters

Task: Filter the data to show only 5-star reviews that also have 5 or more 'cool' votes.

Methods used: The `&` (AND) operator between two conditions.

```
filtered = df[(df['stars'] == 5) & (df['cool'] >= 5)]
display(filtered.head())
```

	business_id	date	review_id	stars	text
6	zp713qNhx8d9KCJJnrw1xA	2010-02-12	riFQ3vxNpP4rWLk_CSri2A	5	Drop wh you're doin and driv here. Afte I.
82	d7_jkx0VPx3uHsUI18iHMQ	2009-06-16	xE2MKvzes5Twg-jfAe8s4Q	5	Beer...c Yoga \n\nLocate in a sma center j.
112	StiQ_lcCY8sX4JI-J6Mufg	2009-11-06	i2WTOTbT1lqu0byxcnY-GA	5	'twas th night befor booyah\nan all the thr.
125	rZbHg4ACfN3iShdsT47WKQ	2010-07-13	hh1L68RTi1FCHNfXGHkc7A	5	That dar Smokehous burger i awesome. star.
144	zp713qNhx8d9KCJJnrw1xA	2010-02-02	qyzPbUwY_QgqdEputNWk6g	5	Perhaps was foolis for me t give L Conde.

✓ Problem 18: Percentiles

Task: Find the value below which 90% of the 'useful' vote counts fall. **Methods used:**

```
quantile(0.9).
```



```
quantile_90 = df['useful'].quantile(0.9)
print(f"90th percentile for useful votes: {quantile_90}")
```

90th percentile for useful votes: 3.0

✓ Problem 19: Unique Counts

Task: Calculate the total number of unique users who wrote reviews in this dataset.

Methods used: `nunique()` on the `user_id` column.

```
unique_users = df['user_id'].nunique()
print(f"Unique Users: {unique_users}")
```

Unique Users: 6403

✓ Problem 20: Slicing and Resetting

Task: Create a new DataFrame of only 1-star reviews and reset the row numbers to start from 0. **Methods used:** `reset_index(drop=True)`.

```
one_star_subset = df[df['stars'] == 1].reset_index(drop=True)
display(one_star_subset.head())
```

	business_id	date	review_id	stars	text
0	IJ0o6b8bJFAbG6MjGfBebQ	2010-09-05	Dx9sfFU6Zn0GYOckijom-g	1	U can go there n check the car out. If u wanna...
1	vvA3fbps4F9nGIAEYKk_sA	2012-05-04	S9OVpXat8k5YwWCn6FAgXg	1	Disgusting! Had a Groupon so my daughter and ...
2	o1GIYYZJjM6nM03fQs_uEQ	2011-11-30	ApKbwpYJdnhhgP4NbjQw2Q	1	I've eaten here many times, but none as bad as...
3	I4vBbCL9QbGiwLuLKwD_bA	2011-11-22	DJVxOfj2Rw9zkIC9tU3i1w	1	I have always been a fan of Burlington's